A Prefix Census of $3n + c$ and $5n + 1$ Cycles, with One Verified Collatz Delay-Record Instance

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Origin Provenance: September 23, 2026 (23:54:15 CDT)
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Abstract

We provide a cryptographically hashed computational census of periodic cycles for the generalized hailstone family $C_c(n) = n/2$ (n even), $3n+c$ (n odd) and $C_5(n) = n/2$ (n even), $5n+1$ (n odd), alongside an exact 1348-step trajectory certificate for the maximal delay-record start below 10^{12}, $n = 989{,}345{,}275{,}647$, reaching 1 with peak $1{,}219{,}624{,}271{,}099{,}764 < 2^{53} - 1$ (SHA-256 trajectory digest f406d4f31af4402276e39a48d0d68e4e587ca0a318913a0b196d5da68332ed87). We prove the sign-conjugacy identity $C_{-c}(n) = -C_c(-n)$, catalog the basin splits of $3n-1$ (349/306/345 across the cycles at $1, 5, 17$ for starts $1 \dots 1000$) and the six cycles of $3n + 5$ (including both length-44 primitive cycles), and align all tables with Belaga–Mignotte and Holden.

1 Main Theorems and Verified Census

Theorem 1 (Verified Delay-Record Instance Below 10^{12})**.** *For $n_0 = 989{,}345{,}275{,}647$, the classical Collatz map C_+ satisfies*

$$C_+^{1348}(989{,}345{,}275{,}647) = 1, \qquad \max_{0 \le k \le 1348} C_+^k(n_0) = 1{,}219{,}624{,}271{,}099{,}764 < 2^{53} - 1.$$

Theorem 2 (Sign Conjugacy)**.** *For every odd integer c and every $n \in \mathbb{Z}$, $C_{-c}(n) = -C_c(-n)$. In particular, the positive cycles of $3n - 1$ at 1, 5, and 17 correspond bijectively to the negative cycles of $3n + 1$ at -1, -5, and -17.*

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